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Geometric Mean Return

The arithmetic mean return assumes that the amount invested at the beginning of each period is the same. In an investment portfolio, however, even if there are no cash flows into or out of the portfolio the base amount changes each year. The previous year's earnings must be added to the beginning value of the subsequent year's investment—these earnings will be "compounded" by the returns earned in that subsequent year. We can use the geometric mean return to account for the compounding of returns.

A geometric mean return provides a more accurate representation of the growth in portfolio value over a given time period than the arithmetic mean return. In general, the geometric mean return is denoted by $\bar{R}_{Gi}$ and given by the following equation for asset i:

$$\bar{R}_{Gi} = \sqrt[T]{(1+R_{i1}) \times (1+R_{i2}) \times \ldots \times (1+R_{i,T-1}) \times (1+R_{iT})} - 1$$

$$= \sqrt[T]{\prod_{t=1}^{T}(1+R_t)} - 1 \,, \tag{3}$$

where R_{it} is the return in period t and T is the total number of periods.

In the example in the previous section, we calculated the arithmetic mean to be 4.00 percent. Using Equation 4, we can calculate the geometric mean return from the same three annual returns:

$$\bar{R}_{Gi} = \sqrt[3]{(1-0.50) \times (1+0.35) \times (1+0.27)} - 1 = -0.0500.$$

Exhibit 2 shows the actual return for each year and the balance at the end of each year using actual returns.

Exhibit 2: Portfolio Value and Performance

	Actual Return for the Year (%)	Year-End Amount	Year-End Amount Using Arithmetic Return of 4%	Year-End Amount Using Geometric Return of –5%
Year 0		EUR1.0000	EUR1.0000	EUR1.0000
Year 1	–50	0.5000	1.0400	0.9500
Year 2	35	0.6750	1.0816	0.9025
Year 3	27	0.8573	1.1249	0.8574

Beginning with an initial investment of EUR1.0000, we will have a balance of EUR0.8573 at the end of the three-year period as shown in the fourth column of Exhibit 2. Note that we compounded the returns because, unless otherwise stated, we earn a return on the balance as of the end of the prior year. That is, we will receive a return of 35 percent in the second year on the balance at the end of the first year, which is only EUR0.5000, not the initial balance of EUR1.0000. Let us compare the balance at the end of the three-year period computed using geometric returns with the balance we would calculate using the 4 percent annual arithmetic mean return from our earlier example. The ending value using the arithmetic mean return is EUR1.1249 (=1.0000 × 1.04^3). This is much larger than the actual balance at the end of Year 3 of EUR0.8573.

In general, the arithmetic return is biased upward unless each of the underlying holding period returns are equal. The bias in arithmetic mean returns is particularly severe if holding period returns are a mix of both positive and negative returns, as in this example.

We will now look at three examples that calculate holding period returns over different time horizons.